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UltraVIDisplays.1.02

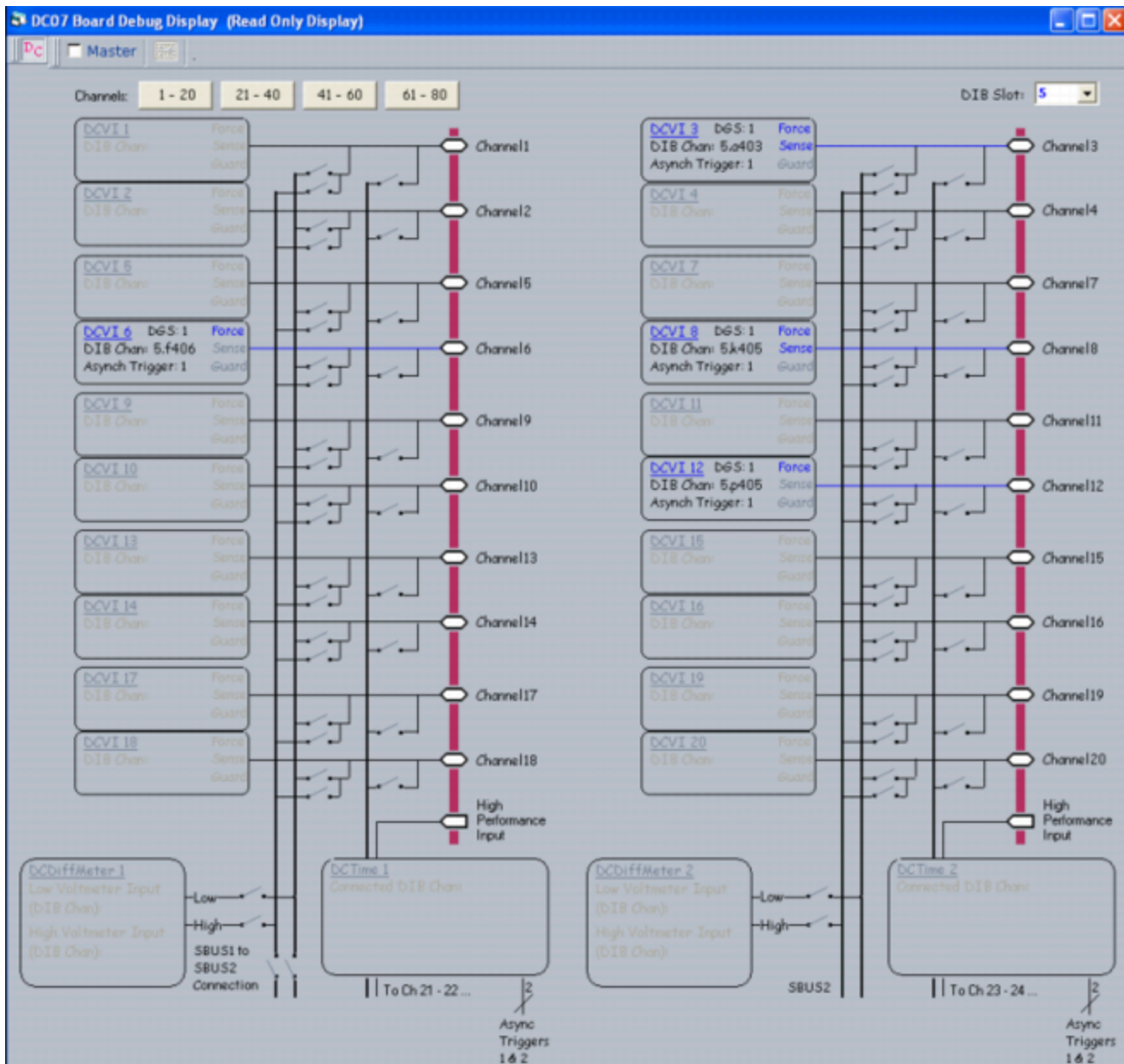


UltraVI80 Debug Displays : UltraVI80 Board Debug Display

UltraVI80 Board Debug Display

The UltraVI80 Board Debug Display is a read-only display showing boardwide connections. The display shows an overview of DCVI, DC Differential Voltmeter, subbus, and DC Time Measurement Unit connections.

To view this display, from the Launch group on the IG-XL Tools tab, select Subsystems>DC07 Board. The display shows all 80 UltraVI80 DCVI channels in groups of 20. The figure shows an example of the display for channels 1 - 20.



UltraVI80 Board Debug Display (showing channels 1 - 20)

Using the Display

As a test is run, the display changes depending on the test program being executed. Channel information is highlighted on each instrument representation and relays open and close as the state of the instrument changes. The signal path is also highlighted using color-coded lines. Click an instrument name to open the debug display for that instrument.

DIB Slot

Use this menu to select a DIB slot to display the instrument board to be debugged. Only DIB slots containing a DC Instrument board appear in this menu.

Channels Buttons

Only 20 channels are displayed in the display at a time. Click the Channels buttons to display each group of 20 channels.

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UltraVDisplays.1.03



UltraVI80 Debug Displays : DCDiffMeter Debug Display

DCDiffMeter Debug Display

The DCDiffMeter Debug Display is an interactive tool that shows the state of a DC Differential Voltmeter. Most parameters are interactive. Features that you can change may be limited to a certain range or selection of options provided in a menu. Noninteractive features are grayed out on the display and cannot be adjusted during debugging.

To view this display, from the Launch group on the IG-XL Tools tab, select Debug Displays>DCDiffMeter. The figure shows an example of the display.

cap_n_s(M) DCDiffMeter Debug Display

Site: 0 ☒ Master

MeterMode: HighSpeed

VoltageRange: 7.00 V

Meter Reading: V

Sample Size: 1

Sample Rate: 8.33 M Hz

HW Avg: 1 ☐ No Strobe

Loop Meter Read Meter

Async Enable: ☐ Async Continuous: ☐

High Pin: cap_n_s
DIB Slot And Channel #4.a303

Low Pin:
DIB Slot And Channel #

Trigger

Asynchronous Trigger

Slope: Positive

Pin:

Chan Type: DCTime

Which: 1

PSet: PSet:
Edit Apply

Capture Waveforms

Resource ID:
Apply

Sample Rate: 8.33 M Hz

Sample Size: 1

Extra Samples: 0

Trigger Stop ClearCapMem

Alarms

Alarm: Over Range Behavior: Force Fail

DCDiffMeter Debug Display

Using the Display

To start debugging, select a pin and site to display the program parameters. Once you select a pin and site, parameters are shown in real time for that instrument.

Relay Graphics

Specifies the state of the DCDiffMeter Connection to the DUT pin. Connect or disconnect relays by clicking them.

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UltraVI80 Debug Displays : DCDiffMeter Debug Display : DCDiffMeter Connections

DCDiffMeter Connections

To connect or disconnect the DCDiffMeter to or from DUT pins using the debug display:

1. [Select a High Pin or Low Pin from the menu.](#)

Selecting a pin displays the DIB slot and channel that the pin is mapped to in the channel map between the two pin menus.

2. [Click relays to connect or disconnect a High or Low Pin.](#)

High Pin	Specifies the list of DCDiffMeter high pins on which the test operates.
Low Pin	Specifies the list of DCDiffMeter low pins on which the test operates.
Relay graphics	Specifies the state of the DCDiffMeter Connection to the DUT pin. Connect or disconnect relays by clicking them.

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UltraVI80 Debug Displays : DCDiffMeter Debug Display : Meter Setup

Meter Setup

You can change DCDiffMeter settings using the debug display. This allows you to change the program while the program is active and being stepped through. Change parameters by selecting items from a menu or clicking a control.

MeterMode	Selects the mode of the DCDiffMeter. Select: - HighAccuracy - HighSpeed (default) See DCDiffMeter Modes.
VoltageRange	Specifies a list of voltage ranges that can be set. A voltage range programed should be higher than the voltage level being measured.
Sample Size	Specifies the number of samples to be read by the meter.
Sample Rate	Specifies the number of samples to be captured per second. This box is grayed out when No Strobe is selected.
HW Avg	Selects the value for Meter hardware averaging. See DCDiffMeter Hardware Averaging.
No Strobe	Select this box to produce a reading based on the current state of the test program but using values stored in memory rather than strobing the signal. This is useful for viewing data while a pattern is running.

	Clear this box to cause the meter to strobe the signal based on the sample size and sample rate programmed. All results are returned to the Avg Measurement box.
Loop Meter	Opens the Meter Loop Tool and displays measurements. This button is grayed out when No Strobe is selected. See Meter Loop Tool.
Read Meter	Makes a DiffMeter reading and displays it in Meter Reading.
Pattern Unlock	Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See PatternInstrumentUnlock in the VBT Syntax Reference.

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UltraVI80 Debug Displays : DCDiffMeter Debug Display : Asynchronous Trigger

Asynchronous Trigger

Allows measurements to be made asynchronous of the test system. A change of state in the Asynchronous Trigger Pin triggers a measurement on the pin connected to the DCDiffMeter.

Asynch Enable	Select this box to have the DCDiffMeter triggered by an asynchronous trigger. If it is not selected, the asynchronous trigger signal is ignored. This box can also indicate that a trigger (or strobe) has occurred when Continuous mode is cleared. (See Asynch Continuous.) See Interaction with the Debug Display.
Asynch Continuous	Select this box to specify that an asynchronous trigger automatically be reenabled after the trigger event. Clear this box to require that an asynchronous trigger be manually reenabled after the trigger event.
Slope	Selects the slope of the asynchronous trigger signal. The slope can be: - Positive - Negative
Pin	Specifies the pin whose signal is used to asynchronously trigger the instrument.
Chan Type	This is the asynchronous trigger pin's channel type. The trigger channel type should match the channel type found in the channel map that

Which

corresponds with the asynchronous trigger pin.
Selects the threshold used to asynchronously trigger the instrument.
Options are:

- 1: threshold 1
- 2: threshold 2

For more information, see [Onboard Asynchronous Triggering of a DCDiffMeter](#).

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UltraVI80 Debug Displays : DCDiffMeter Debug Display : Capture Waveforms

Capture Waveforms

The meter stores a number of samples based on input in memory to capture the signal. The capture memory maximum depth is 4M samples. Capture Waveforms is only available if MeterMode is set to HighSpeed. The memory depth is 4M. A signal is captured based on the settings specified below.

Resource ID	Specifies a signal name to capture. Select the signal from a list of preprogrammed signals.
Apply	Loads the specified Resource ID.
Sample Rate	Specifies the number of samples to be captured per second.
Sample Size	Specifies the number of samples specified to be read by the meter.
Extra Samples	Specifies how many extra samples should be taken in the capture. For example, if the Sample Size is 3 and Extra Samples is set to 5, then the total number of samples taken is 8.
Trigger	Triggers the instrument to start the capture and opens the WaveScope.
Stop	Stops the signal capture.
ClearCapMem	Clears the capture memory.

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UltraVI80 Debug Displays : DCDiffMeter Debug Display : PSets

PSets

The PSet controls allow you to select an available PSet and optionally apply the settings to the DCVI hardware. You can use the following controls:

PSet	Select an available PSet. The list contains all PSets defined for the current pin.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Apply	Applies the selected PSet to the hardware for that channel.

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UltraVI80 Debug Displays : DCDiffMeter Debug Display : Alarms

Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging. You can set the following parameters during normal debug procedures:

Alarm	Selects all alarms or a specific alarm to set an action for. Selecting an alarm only affects that alarm for that pin.
Behavior	Selects an action for the alarm or alarms selected in the Alarm menu. Alarm actions specify what should occur if the selected alarm condition exists for the pin. This menu is grayed out until you select from the Alarm menu. Options are: - Force Bin - Off - Force Fail - Default - Continue - Raise Error

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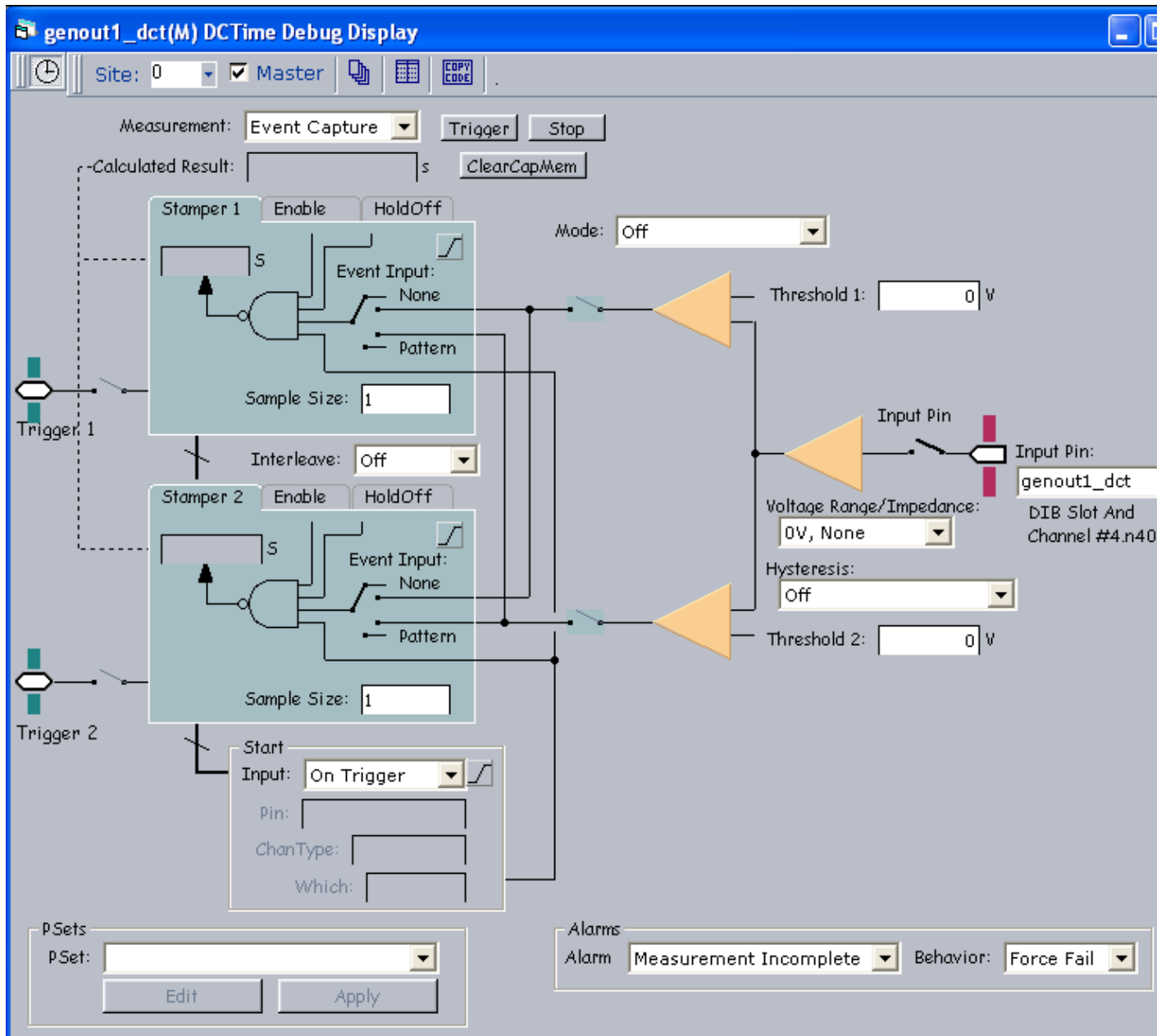


UltraVI80 Debug Displays : DCTime Debug Display

DCTime Debug Display

The DCTime Debug Display is an interactive tool that shows the state of DC Time Measurement Unit. Most parameters are interactive. Features that you can change may be limited to a certain range or selection of options provided in a menu. Noninteractive features are grayed out on the display and cannot be adjusted during debugging.

To view this display, from the Launch group on the IG-XL Tools tab, select Debug Displays>DCTime. The figure shows an example of the display.



DCTime Debug Display

Using the Display

To start debugging, select a pin and a site to display the program values. Select sites and pins from the menu. Once you select a pin and a site, parameters are shown for that instrument.



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UltraVI80 Debug Displays : [DCTime Debug Display](#) : DCTime Connections

DCTime Connections

Connect or disconnect the DCTime from DUT pins with the debug display using the following procedure:

- | | |
|----|---|
| 1. | Select a pin from the menu. |
|----|---|

Selecting a pin displays the DIB slot and channel that the pin is mapped to below the [Pin](#) menu.

- | | |
|----|---|
| 2. | Click the Input Pin relay to connect or disconnect the sense lines. |
|----|---|

Pin	Specifies the list of DCTime pins on which the test operates.
Input Pin	Specifies the state of the DCTime connection to the DUT pin. Change the relay state to connect or disconnect by clicking it.

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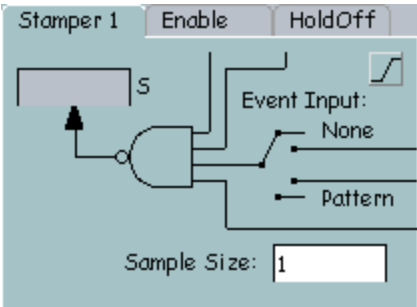
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UltraVI80 Debug Displays : DCTime Debug Display : Stamper Tab

Stamper Tab

Click a Stamper tab to bring its contents to the front. Choose the tab for stamper 1 or stamper 2.



DCTime Stamper Tab

You can change the following controls:

Measurement results field	Displays the measurement result for the stamper.
Sample Size	Specifies the number of samples to be read. Sample size is 1 - 512 samples.
Event Input	The event on which the time stamper is to stamp on. Click the relay to change the connection to one of the following: - None - Threshold 1 - Threshold 2

- [Pattern](#)

Event Slope

[See Event Signal.](#)

The negative or positive slope. The icon’s picture displays either a negative or positive slope. Slope does not apply when the stamper is Off or when the input is a pattern.

[See Event Signal.](#)

			
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UltraVI80 Debug Displays : DCTime Debug Display : Enable Tab

Enable Tab

Click an Enable tab to bring its contents to the front. Choose the tab for stamper 1 or stamper 2.

Stamper 1 Enable HoldOff

Input: None

Pin:

ChanType:

Which:

Mode: Gate Window

DCTime Enable Tab

You can change the following controls:

Input	Selects the event on which the stamper becomes enabled. Options are: - None - Pin - Pattern If Pin is selected, the Enable Pin field is required. Otherwise, the Enable Pin field is not required and is grayed out.
Pin (enable)	Selects a programmed pin. This is the pin that enables the time stamper. This text box is only active if the Input menu is set to Pin.

ChanType (enable pin)	The enable pin's channel type. The enable channel type should match the channel type found in the specified channel map. This text box is only active if the Input menu is set to Pin.
Which (enable pin)	Selects threshold 1 or threshold 2 to asynchronously enable the instrument. Options are: 1: threshold 1 2: threshold 2 This text box is only active if the Input menu is set to Pin.
Mode	Selects how events are captured whenever the enable input is another pin's signal. Options are: - Gate Window: Stamper captures events as long as its enable input is at gate_high. The stamper cannot stamp if enable input is gate_low. - Trigger Each Event: Stamper must receive a trigger on its enable input before capturing the next event signal. See Enable Mode.
Slope (enable)	Sets negative or positive slope. The icon's picture displays either a negative or positive slope. Slope does not apply when the stamper is Off or when the input is a pattern.

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UltraVI80 Debug Displays : DCTime Debug Display : HoldOff Tab

HoldOff Tab

Click the HoldOff tab for stamper 1 or stamper 2 to bring its contents to the front.

Stamper 1	Enable	HoldOff
Mode: Off		
Count: 0		
Time: 0 s		

DCTime HoldOff Tab

You can change the following controls:

Mode

Select which hold-off is used to allow time for a signal to settle before making a time measurement. Options are:

- **Precount Once:** Counts a programmed amount of time or counts a programmed number of events before making the first time stamp on the first sample only.
- **Before Each:** Holds off a programmed amount of time or counts a programmed number of events before making each and every stamp.
- **Off:** No delay.

	See Hold-Off Mode.
Count	Displays how many events to hold off before a signal is stamped. Hold off count waits for a certain number of events to occur before making a time measurement. See Hold-Off Count.
Time	Displays the set time before a signal is stamped. This method waits at a minimum the specified amount of time before a time measurement is be made. See Hold-Off Time.

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UltraVI80 Debug Displays : DCTime Debug Display : Start

Start

This area of the displays controls the start signal for the DCTime.

Input	Selects the event a start responds to. Options are: - On Trigger - Pin A start is equivalent to an on/off switch for the time stamper. If the stamper has received a start signal, then the stampers are on. Otherwise, they are off. If Pin is selected, the Pin field is required. Otherwise, the Pin field is not required and is grayed out. See Start Modes.
Pin (start)	Selects from the programmed pins. This is the pin of the Start signal. This text box is only active if the Input menu is set to Pin.
ChanType (start pin)	The start pin’s channel type. It should match the channel type found in the channel map that corresponds with the start pin. This box is only active if the Input menu is set to Pin.
Which (start pin)	Selects if threshold 1 or threshold 2 is used to asynchronously start the instrument. Options are:

- 1: threshold 1
- 2: threshold 2

This box is only active if the Input menu is set to Pin.

Slope (start pin) Sets negative or positive slope. The icon's picture displays either a negative or positive slope. Slope does not apply when the stamper is off or when the input is a pattern.

			
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[UltraVI80 Debug Displays](#) : [DCTime Debug Display](#) : [Stamper Setup](#)

Stamper Setup

Interleave

Options are:

- [On](#): Stamper 2 only captures events after Stamper 1 has captured to avoid misalignments of the time stamps.
- [Off](#): Interleave mode is off and stampers can stamp in any order.

[See Interleave Mode.](#)

Measurement

Displays the result of the time measurement for the stamper. This measurement is based on the selected measurement type. Each time you [click Trigger](#), you must select the correct measurement type. Options are:

- [Fall Time](#)
- [Rise Time](#)
- [Frequency](#)
- [Period](#)
- [Duty Cycle](#)

- [Pulse Width](#)
- [Event Capture](#)

[See Measurement Types.](#)

Trigger	Initiates the instrument to start a capture.
Stop	Stops the signal capture.
ClearCapMem	Clears the capture memory.
Calculated Result	Displays the result of the DCTime capture based on the Measurement.

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Front End Schematic

Threshold1/2	Specifies the value at which DCTime should either stamp or trigger.
Hysteresis	Removes noise around the threshold crossing. The gain and offset calibration constants with hysteresis turned on depend on whether you are looking for the positive or negative crossing. These constants are adjusted automatically when hysteresis type is changed. Options are: • Off: Hysteresis is off. • On Positive Adjust: Automatically adjusts voltage threshold for a positive slope. • On Negative Adjust: Automatically adjusts voltage threshold for a negative slope. • On: Hysteresis is on. You must take into account the shift in the voltage threshold. See Hysteresis.
Voltage Range/Impedance	Sets the voltage range and the impedance of the incoming signal so that it can be attenuated. Select a setting from the menu.

	See Voltage Range and Impedance .
Mode	Determines the connection of the front end. Options are: • Off: No connections made. • Stamper: Connection to front end is made. • On Board Trigger: Connection to front end and trigger is made.
Pattern Unlock	Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See PatternInstrumentUnlock in the VBT Syntax Reference.

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UltraVI80 Debug Displays : DCTime Debug Display : PSets

PSets

The PSet controls allow you to select an available PSet and optionally apply the settings to the DCVI hardware. You can use the following controls:

PSet	Select an available PSet. The list contains all PSets defined for the current pin.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Apply	Applies the selected PSet to the hardware for that channel. This control is grayed out if no PSets are available.

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UltraVI80 Debug Displays : DCTime Debug Display : Alarms

Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging.
You can set the following parameters during normal debug procedures:

Alarm	Selects all alarms or a specific alarm to set an action for. Selecting an alarm only effects that alarm for that pin. See DCTime Alarms.
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Behavior

Selects an action for the alarm or alarms selected in the Alarm menu. Alarm actions specify what should occur if the selected alarm condition exists for the pin. This menu is grayed out until you select from the Alarm menu. Options are:

- Force Bin
- Off
- Force Fail
- Default
- Continue
- Raise Error

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UltraVIDisplays.1.20



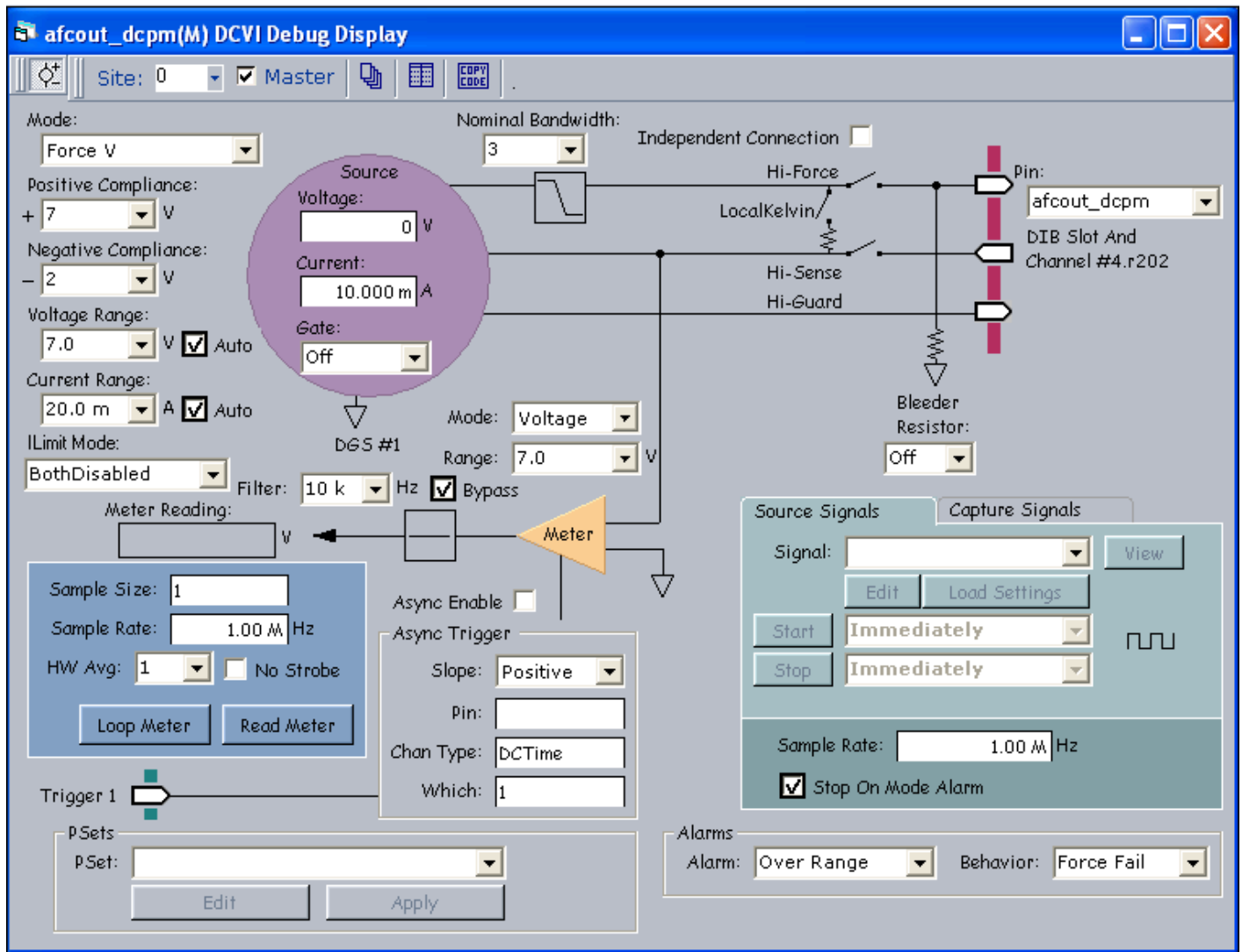
UltraVI80 Debug Displays : DCVI Debug Display

DCVI Debug Display

The DCVI Debug Display is an interactive tool that shows the state of a selected DCVI channel. Features that you can change may be limited to a certain range or selection of options provided in a menu. Non-interactive features are grayed out and cannot be adjusted.

To view this display, from the Launch group on the IG-XL Tools tab, select Debug Displays>DCVI.

The figure shows an example of the display when an UltraVI80 pin is selected.



DCVI Debug Display

Using the Display

To begin debugging, select a pin and site to display parameters and make changes. Select a site and a pin from the menus.

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UltraVI80 Debug Displays : [DCVI Debug Display](#) : DCVI Connections

DCVI Connections

Connect or disconnect the DCVI from DUT pins with the debug display using the following procedure:

1. [Select a pin from the menu.](#)

Selecting a pin displays the DIB slot and channel that the pin is mapped to in the channel below the Pin menu.

2. [Click the relays to connect or disconnect the force and sense lines.](#)

Pin	The list of DCVI pins on which the test operates. All DCVI pins are displayed in the menu. Choosing a pin determines if the debug display is showing an UltraVI80 instrument.
Independent Connection	Allows you to make connections with the force and sense lines independent from each other. With the check box selected, each connection relay closes independently from all others. This is not true when the check box is cleared.
Relay graphics	The state of the DCVI connection to the DUT pin. Click a relay to connect or disconnect it.
Bleeder Resistor	Allows you to connect the DCVI bleeder resistor to help discharge DIB capacitance. Select the following:

- [Off](#): Bleeder resistor disconnected.
- [On](#): Bleeder resistor connected.
- [Auto](#): Bleeder resistor is connected if force relay is opened and disconnected if force relay is closed. Default.

[See DCVI Bleeder Resistor.](#)

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UltraVI80 Debug Displays : DCVI Debug Display : DCVI Setup

DCVI Setup

The DCVI Schematic section displays the current programmed settings of the selected DCVI. You can change program parameters by entering a value in a text box or selecting from a menu. You can adjust the following parameters during normal debug procedures:

Positive Compliance	Select or enter a positive compliance range. Since the UltraVI80 has only one asymmetrical range of -2V to +7V, and this parameter cannot really be adjusted. See Compliance.
Negative Compliance	Select or enter a negative compliance range. Since the UltraVI80 has only one asymmetrical range of -2V to +7V, and this parameter cannot really be adjusted. See Compliance.
Voltage Range	The voltage range. For UltraVI80, only one range is available. Selecting Auto allows IG-XL to handle setting the voltage range automatically. Clearing Auto requires a manual entry. See Force Voltage.
Current Range	Select a current range from the menu. Selecting Auto allows IG-XL to handle setting the current range automatically. Clearing Auto requires a manual entry. See Force Current.

Voltage	The programmed voltage. Entering a valid value programs the DCVI to that value. You may need to change the voltage range when entering a new voltage.
Current	The programmed current. Entering a valid value programs the DCVI to that value. You may need to change the current range when entering a new current.
Gate	Displays the state of the gate. Select from the following: • Off (default) • On • OffHiZ See Gate Function.
Mode	Displays the currently programmed mode, and allows you to change the mode. The DCVI must be gated off to change mode. Select a DCVI mode from the following: • Force V: See Force Voltage • Force I: See Force Current • High Impedance: See High Impedance Mode
Nominal Bandwidth	Displays the current programmed bandwidth setting and allows the bandwidth to be programmed to one of the selections in the menu. See DCVI Bandwidth Compensation Settings.
Pattern Unlock	Allows you to use select and monitor pattern pins while that pattern is running (that is, in a CPU loop). See PatternInstrumentUnlock in the VBT Syntax Reference.



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UltraVI80 Debug Displays : DCVI Debug Display : DCVI Meter Setup

DCVI Meter Setup

Each DCVI channel has a dedicated meter. Use the meter to make measurements on DUT pins through proper relay connections and isolations. The channel meters can strobe the measurement asynchronously by communicating with the onboard trigger lines. These lines are driven by the outputs of the DCTime front-end circuits.

You can adjust the following parameters during normal debug procedures:

Mode	Specifies the mode of operation of the DCVI meter as one of the following: - Voltage - Current
Range	Specifies the metering range of the DCVI meter. This menu switches between current and voltage based on the Mode selected. See DCVI Meter.
Filter	Specifies the filter setting for the DCVI meter. Select from the menu: 10 k 100 k See Metering Filter.

Bypass	Select this box to set the meter filter to bypass. Clear this box to use the filter setting in the Filter menu.
Sample Size	Specifies the number of samples to be read by the meter. See DCVI Meter .
Sample Rate	Specifies the number of samples to be captured per second. This box is grayed out when No Strobe is selected. See DCVI Meter .
HW Avg	Selects the value for meter hardware averaging.
No Strobe	Select this box to produce a reading based on the current state of the test program but use values stored in memory rather than strobing the signal. This is useful in viewing data while a pattern is running. Clear this box to cause the meter to strobe the signal based on the sample size and sample rate programmed. See DCVI Meter .
Loop Meter	Opens the Meter Loop Tool and displays measurements as long as the meter is making measurements. This button is grayed out when the No Strobe is selected. See Meter Loop Tool .
Read Meter	Makes a DCVI meter reading and displays it in Meter Reading label.

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UltraVI80 Debug Displays : DCVI Debug Display : Source Signals

Source Signals

A signal can be sourced to a DCVI pin. Enabling Autoload allows IG-XL to enter a preprogrammed signal automatically and disables the Load and Unload buttons. If Autoload is cleared, you must manually load a signal or an error occurs. The source memory maximum depth is 512K.

Signal	The menu includes the names of the defined signals. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the sourced signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be sourced. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Load Settings	Specifies If the driver computes and loads the register settings needed to create the signal named in the Signal menu.
Start	Starts the signal based on the setting specified in the menu next to the button:

- **Immediately**: The signal starts and loops automatically each time it ends.
- **Once**: The signal is sourced and stops at the end.

The square wave is a sourcing indicator that moves when the source is sourcing on the tester. This feature only works on line.

Stop

Stops the signal based on the setting specified in the menu next to the button:

- **Immediately**: The signal terminates at that point.
- **At End**: The signal stops at the end of the current cycle.

Sample Rate

Specifies the number of samples to be sourced per second.

Stop On Mode Alarm

Select this box to terminate signal sourcing if an alarm occurs. If unselected, the program continues based on the alarm behavior setting.

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UltraVI80 Debug Displays : DCVI Debug Display : Capture Signals

Capture Signals

The meter stores a number of samples based on input in memory to capture the signal. The capture memory maximum depth is 512K samples for signals. The strobe memory maximum depth is 1K.

Signal	The menu includes the names of the defined signals. If a default signal has been defined, the default signal’s name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Brings up the WaveScope tool and plots the captured signal segment on the WaveScope display. This allows you to visually preview and analyze the signal that is to be captured. When you are running the program on a test system, WaveScope displays the samples and scales them by the amplitude. When working offline, the WaveScope displays the ideal waveform.
Load Settings	Specifies if the driver computes and loads the register settings needed to create the signal named in the Signal menu.
Trigger	Triggers the instrument to start the capture.
Stop	Stops the signal capture.
Sample Rate	Specifies the number of samples to be captured per second.
Sample Size	Specifies the number of samples specified to be read by the meter.

Extra Samples	Specifies how many extra samples to take in a capture. For example, if Sample Size is 3 and Extra Samples is set to 5, then the total number of samples taken is 8.
ClearCapMem	Clears the capture memory.

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UltraVI80 Debug Displays : DCVI Debug Display : Asynchronous Trigger

Asynchronous Trigger

Allows measurements to be made asynchronous of the test system. A change of state in the Asynchronous Trigger Pin triggers a measurement on the pin connected to the DCVI meter.

Asynch Enable	Select this box to have the DCVI Meter triggered by an asynchronous trigger. If it is not selected, the asynchronous trigger signal is ignored. This box can also indicate that a trigger (or strobe) has occurred when Continuous mode is not enabled. See Interaction with the Debug Display.
Slope	Select the slope of the asynchronous trigger signal. Options are: - Positive - Negative
Pin	Specifies the pin whose signal is used to asynchronously trigger the instrument.
Chan Type	The asynchronous trigger pin's channel type. The trigger channel type should match the channel type found in the channel map that corresponds with the asynchronous trigger pin.
Which	The threshold used to asynchronously trigger the instrument. Options are:

- 1: threshold 1
- 2: threshold 2

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UltraVI80 Debug Displays : DCVI Debug Display : PSets

PSets

The PSet controls allow you to select an available PSet and optionally apply the settings to the DCVI hardware. You can use the following controls:

PSet	Select an available PSet. The list contains all PSets defined for the current pin.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. This control is grayed out if no PSets are available. See PSet and Signals Editor.
Apply	Applies the selected PSet to the hardware for that channel. This control is grayed out if no PSets are available.

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UltraVI80 Debug Displays : DCVI Debug Display : Alarms

Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging. You can set the following parameters during normal debug procedures:

Alarm	Select all alarms or a specific alarm to set an action for. Selecting an alarm only effects that alarm for that pin. See DCVI Meter
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Behavior

Select an action for the alarm or alarms selected in the Alarm menu. Alarm actions specify what should occur if the selected alarm condition exists for the pin. This menu is grayed out until you select from the Alarm menu. Options are:

- Force Bin
- Off
- Force Fail
- Default
- Continue
- Raise Error

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UltraVI80 Debug Displays : Meter Loop Tool

Meter Loop Tool

The Meter Loop Tool is a standard IG-XL tool that allows you to loop a meter without disabling the debug display from which you started looping. You can also:

- [Loop multiple channels](#)
- [Monitor all results in one place](#)
- [Start or stop all channels simultaneously](#)
- [Start WaveScope for any pin](#)

For more information, see [Meter Loop Tool](#).

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UltraVI80 Debug Displays

UltraVI80 Debug Displays

The UltraVI80 instrument debug displays are graphical interfaces used to assist with troubleshooting a program. A program must be validated before you can use debug displays to debug a program.

The following topics are included:

- [UltraVI80 Board Debug Display](#)
- [DCDiffMeter Debug Display](#)
- [DCTime Debug Display](#)
- [DCVI Debug Display](#)
- [Meter Loop Tool](#)

This section describes features specific to the UltraVI80 displays. For information on using DC30 and DC75 displays, see DC Debug Displays.

For more information, see:

- [Using Debug Displays in TDE](#)
- [Producing Code from Debug Displays](#)

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